CLINICAL · LARYNX & AIRWAY

Foreign bodies of the airway

A toddler with a peanut in his mouth laughs, gasps and coughs; the coughing stops, and for days or weeks he seems well except for a wheeze that is treated as asthma. Inhaled foreign bodies are a leading cause of accidental death in small children, and a missed one slowly destroys a lung. Suspect it from the story, look for the triad of cough, wheeze and reduced air entry, do not trust a normal chest film, and remember the rule of airway foreign bodies: a negative bronchoscopy is far better than a missed foreign body.

AT A GLANCE

- Choking history is enough.
- A normal film doesn't exclude it.
- Rigid bronchoscopy is the treatment.

1 CHAPTER ONE Overview

Who, what and where.

1.1 Who

Children, more than half under four, who inhale suddenly while laughing, crying, playing or running with something in the mouth; infants have no molars to grind nuts and a weak cough. In adults, aspiration occurs in coma, sleep, intoxication, seizures, under anaesthesia (teeth, dentures), with neurological disease, and in the elderly.

1.2 What

- **Vegetable (organic):** peanuts above all, also seeds (watermelon, pumpkin), beans, peas, grains, nuts, pieces of carrot or apple; regional patterns follow diet (areca nut in parts of India, watermelon seeds in Egypt). They absorb water and swell, and their oils provoke an intense *vegetal bronchitis* with oedema, granulation and suppuration.

- **Non-vegetable:** plastic toys and whistles, beads, pins and needles, nails, wires, ball bearings, coins, pen caps and springs; inert ones may stay silent for long periods.
- **Food boluses** that obstruct the larynx in adults (the "café coronary").

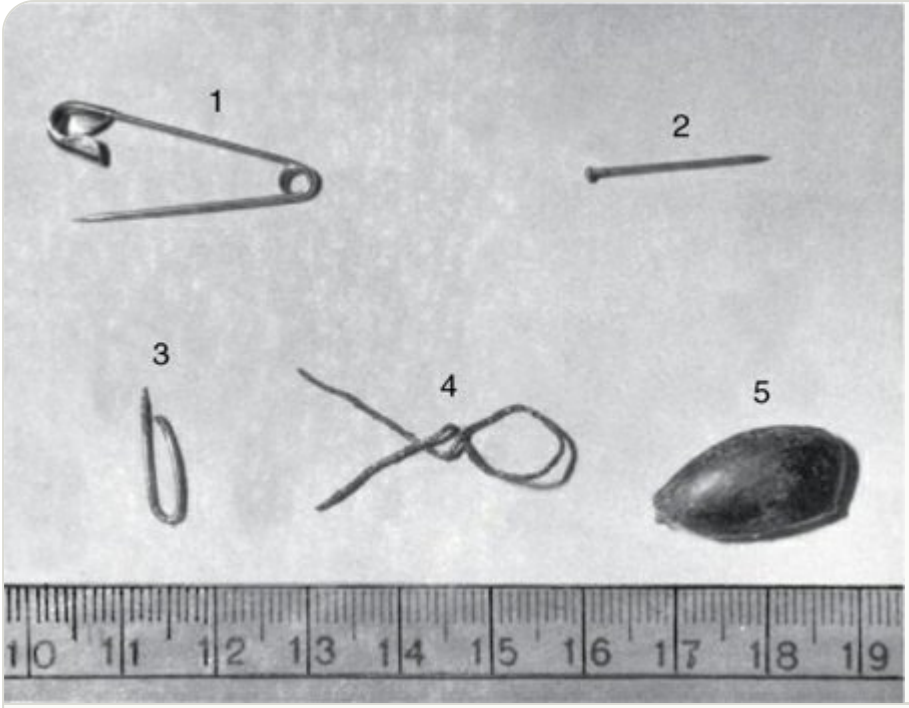


FIGURE 1 Laryngeal foreign bodies: (1) safety pin, (2) pin, (3, 4) twisted wire, (5) a fruit seed. Dhingra, fig. 65.1

1.3 Where

Large objects lodge above the glottis; small ones pass to the trachea and bronchi, most often into the *right main bronchus*, which is wider, shorter and more vertical than the left (in young children the difference is less). Sharp objects stick anywhere.

2 CHAPTER TWO Clinical features

Three stages.

2.1 Stage 1: the event

Sudden choking, gagging, coughing and wheezing, sometimes cyanosis, lasting minutes. The object may be coughed out or lodge. Often the event is not witnessed, or forgotten.

2.2 Stage 2: the silent interval

The mucosa adapts and the cough settles, for hours to weeks, depending on the object.

2.3 Stage 3: complications

Obstruction, inflammation and infection, according to the site.

FEATURES BY SITE

SITE	FEATURES
Larynx	Complete obstruction: aphonia, inability to cough or breathe, rapid death. Partial: pain, hoarseness, croupy cough, stridor, dyspnoea, haemoptysis
Trachea	Choking, cough, biphasic stridor and wheeze heard at the open mouth; a mobile object gives an *audible slap* and *palpable thud* as it strikes the undersurface of the folds
Bronchus	The triad of *cough, wheeze and reduced air entry* on one side; later collapse, emphysema, pneumonia, bronchiectasis, lung abscess

2.4 Patterns of bronchial obstruction

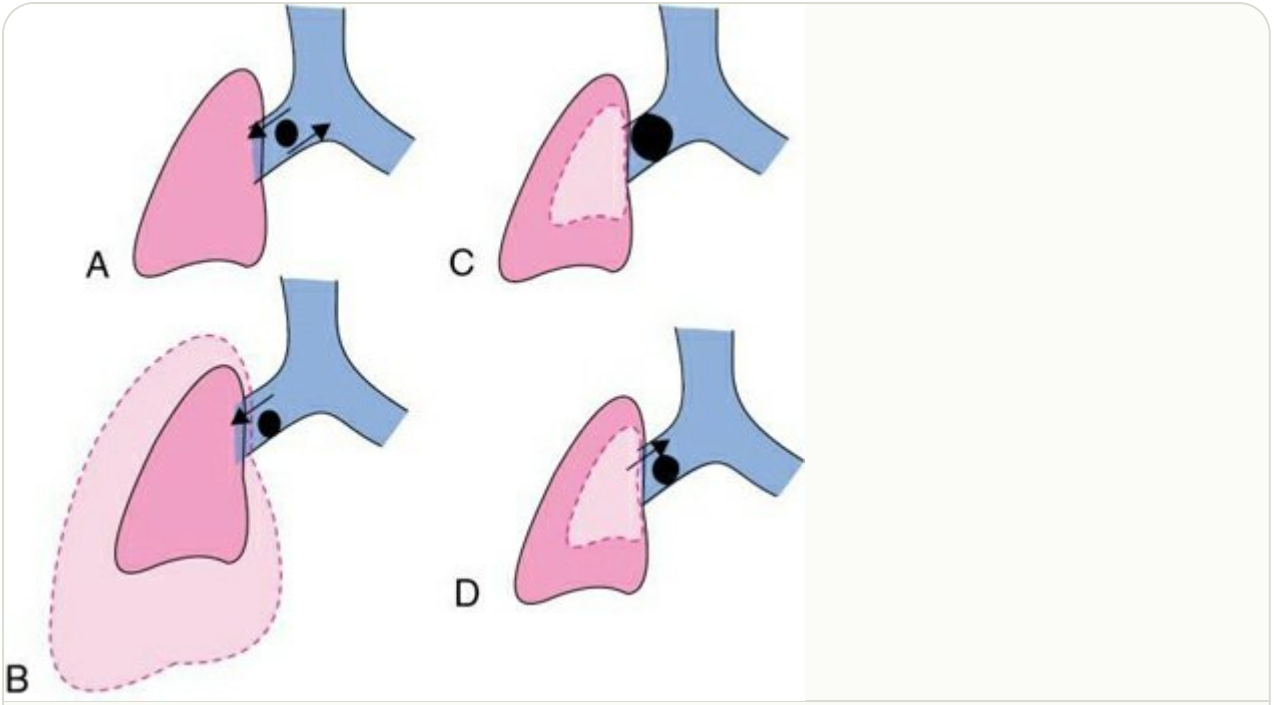


FIGURE 2 Types of bronchial obstruction: (A) partial (bypass valve) with wheeze; (B) check-valve with obstructive emphysema; (C) complete (stop valve) with collapse; (D) ball valve.

Dhingra, fig. 65.2

- **Bypass valve:** air passes both ways; only a wheeze.
- **Check valve:** air enters but cannot leave; *obstructive emphysema* of the lobe or lung, with hyper-resonance, reduced breath sounds and mediastinal shift to the other side.
- **Stop valve:** complete obstruction; air is absorbed and the segment collapses (*atelectasis*), with dullness and shift towards the affected side.
- **Ball valve:** a mobile object enters on inspiration but blocks on expiration.

An emphysematous bulla may rupture into a pneumothorax, and a mobile object may shift sides, changing the signs.

THE ASTHMATIC CHILD WHO IS NOT ASTHMATIC

New wheeze that is one-sided, does not respond to bronchodilators, or follows a choking episode; recurrent pneumonia in the same lobe; a persistent cough after eating nuts: think foreign body and arrange bronchoscopy.

3 CHAPTER THREE Diagnosis

History, examination, imaging.

3.1 History

The single most important clue: a witnessed choking episode has high sensitivity. Ask directly, and ask other family members.

3.2 Examination

Respiratory rate and effort, stridor, voice, unilateral wheeze, reduced air entry, signs of collapse or hyperinflation, and the audible slap of a tracheal object.

3.3 Imaging

- **Soft tissue radiographs of the neck (anteroposterior and lateral, neck extended):** radio-opaque objects, sometimes radiolucent ones in the larynx and trachea. A coin in the trachea lies edge-on in the PA view (sagittal), whereas one in the oesophagus lies flat (coronal).
- **Chest radiograph (PA and lateral):** the object if radio-opaque; lobar or segmental *collapse*; unilateral *hyperinflation* with mediastinal shift away; pneumothorax, pneumomediastinum; pneumonia or bronchiectasis in late cases.
- **Inspiratory and expiratory films, or fluoroscopy** (lateral decubitus films in young children): air trapping on the affected side, which does not deflate on expiration, and the mediastinum swings towards the normal side on expiration.
- **CT of the chest** (low dose) in doubtful, stable cases; virtual bronchoscopy.

A NORMAL CHEST FILM DOES NOT EXCLUDE IT

Most inhaled foreign bodies in children are radiolucent nuts and seeds, and the film is normal in a large proportion, especially in the first day. If the story is suggestive, proceed to bronchoscopy.

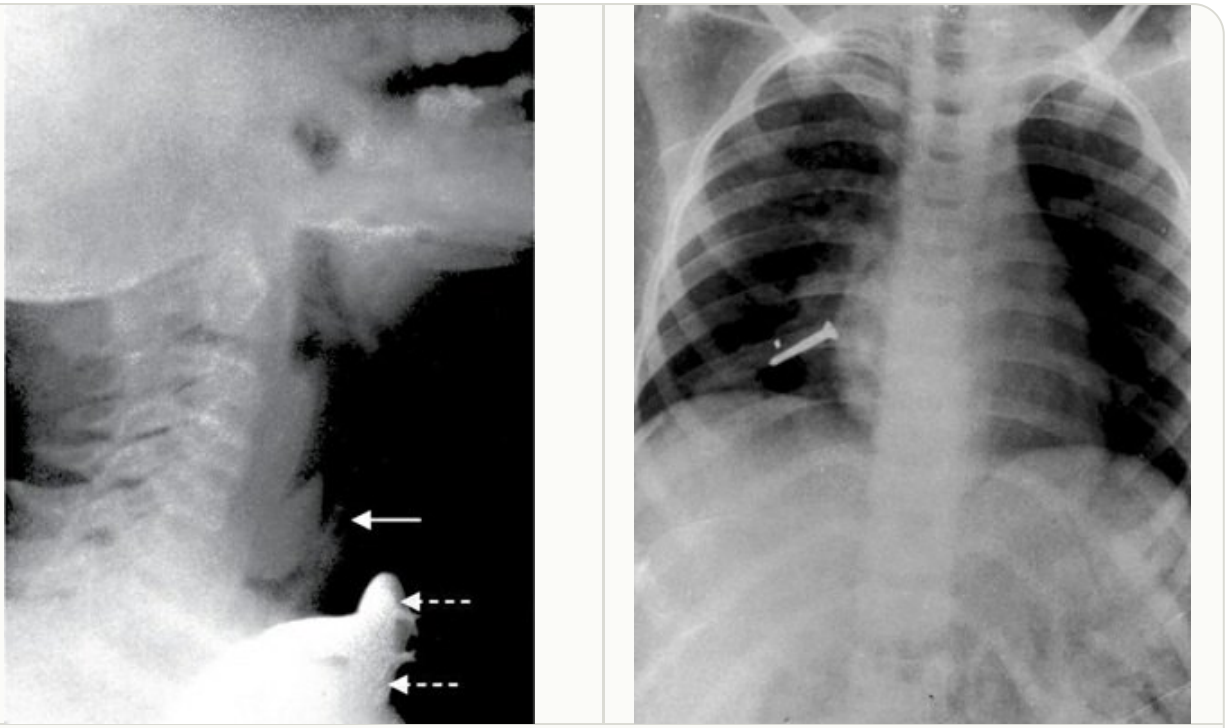


FIGURE 3 First, a radiolucent fruit seed (chiku) in the subglottis (arrow) in a child with a tracheostomy. Second, a nail in the right bronchus on a PA chest film.

Dhingra, figs 65.3 and 65.4



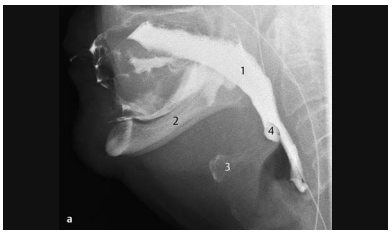
FIGURE 4 First, a broken piece of a Fuller's tracheostomy tube in the left bronchus. Second, (A) a spring from a ball pen in the right bronchus, and (B) the spring after removal.

Dhingra, figs 65.5 and 65.6

IMAGING IN THE APP 2 cases

Foreign bodies and aspiration on imaging.





Foreign Bodies
Trauma · 3 images

Aspiration
Infection & inflammation · 3 images

4 CHAPTER FOUR Management

First aid, then the bronchoscope.

4.1 Complete obstruction: first aid

A patient who cannot speak, breathe or cough needs immediate action:

- **Infants under one year:** five back blows with the infant prone along the forearm, head down, then five chest thrusts; repeat. No abdominal thrusts.
- **Children and adults:** five back blows, then five *abdominal thrusts (Heimlich manoeuvre)*: standing behind, fist below the xiphisternum, sharp inward and upward thrusts that use the residual lung air to expel the object; repeat.
- **If unconscious:** start cardiopulmonary resuscitation; look in the mouth and remove a visible object (no blind finger sweeps).
- If these fail and expertise is available: laryngoscopy and removal with Magill forceps, or cricothyroidotomy (needle in small children) or emergency tracheostomy.



FIGURE 5 The Heimlich manoeuvre: a sudden thrust upwards and backwards below the epigastrium squeezes air from the lungs to dislodge the object.

Dhingra, fig. 65.7

PARTIAL OBSTRUCTION

A patient who can cough and breathe should be encouraged to keep coughing and taken calmly to theatre; back blows and thrusts may convert a partial into a complete obstruction.

4.2 Laryngeal foreign bodies

After the emergency, removal by direct laryngoscopy under general anaesthesia, or by laryngofissure for an impacted object; tracheostomy first if the airway is compromised.

4.3 Tracheobronchial foreign bodies

Removed by *rigid ventilating bronchoscopy* under general anaesthesia, ideally with spontaneous ventilation, by a prepared team: an anaesthetist and surgeon who agree a plan, bronchoscopes of the right size and one size either side, telescopes and *optical forceps*, two laryngoscopes, a range of forceps, Dormia baskets and

Fogarty balloon catheters, and suction, all checked beforehand. Practise grasping a duplicate object before starting.

- Urgent removal is needed for airway obstruction and for vegetable matter that will swell; otherwise it is done on the next planned list with a full team, not in the middle of the night.
- Small objects are withdrawn through the scope; larger ones are held against the tip and the scope, forceps and object removed together; a sharp point is sheathed in the scope or turned to trail.
- Steroids reduce oedema; physiotherapy and antibiotics treat secondary infection.
- Fluoroscopy (C-arm) for peripheral metallic objects; flexible bronchoscopy in selected adults; tracheostomy and bronchoscopy through the stoma for large objects; *thoracotomy and bronchotomy* for objects that cannot be reached.

OPERATIVE GUIDES IN THEATRE 2 operations

Tracheoscopy, bronchoscopy and airway foreign bodies.

Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies
Airway foreign body
Bronchoscopy
Bronchoscopy

4.4 Complications of removal

Oedema of the subglottis, bronchospasm, bleeding from granulations, pneumothorax and pneumomediastinum, fragmentation of nuts with distal spread, and complete obstruction if the object is dropped in the glottis: push it back into a main bronchus and ventilate the other lung, then retrieve it.

4.5 Late complications of a retained object

Recurrent pneumonia, bronchiectasis, lung abscess, granulation tissue and stenosis.

4.6 Prevention

No nuts, popcorn, hard sweets or whole grapes for children under four or five; supervise meals and play; age-appropriate toys without small parts; never hold pins

or small objects in the mouth.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Three stages** of symptoms and the silent interval.
- **Why the right bronchus.**
- **Signs by site:** audible slap and palpable thud; the triad.
- **Valve types** and their radiological signs; inspiratory and expiratory films.
- **First aid:** back blows, chest thrusts in infants, Heimlich in older children and adults.
- **Rigid bronchoscopy:** equipment and technique.

5.2 Common pitfalls

COMMON PITFALLS

- Excluding a foreign body because the chest film is normal.
- Treating a sudden one-sided wheeze in a toddler as asthma.
- Performing back blows or the Heimlich manoeuvre on a patient who is coughing effectively.
- Attempting removal without the right-sized instruments checked and ready.

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IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Aspiration

- Occasionally aspiration is caused by mechanical failure of the epiglottis to close because of the presence of tumor, but most cases are due to neurologic (sensory) disorders without any visible pathology apart from the aspiration. Beware of aspiration pneumonia.
- The above-mentioned pathologies are partly demonstrated in other sections.



First stage: the barium contrast (1) just reaches the esophagus. For orientation note the mandible (2), hyoid (3), and vallecula (4)….